

Setup visual studio 2022 and Web application with ASP.NET MVC 6.0



.NET
Core

In this tutorial, I'll show you how to download and install Visual Studio 2022 and construct a simple.NET Console application step by step.

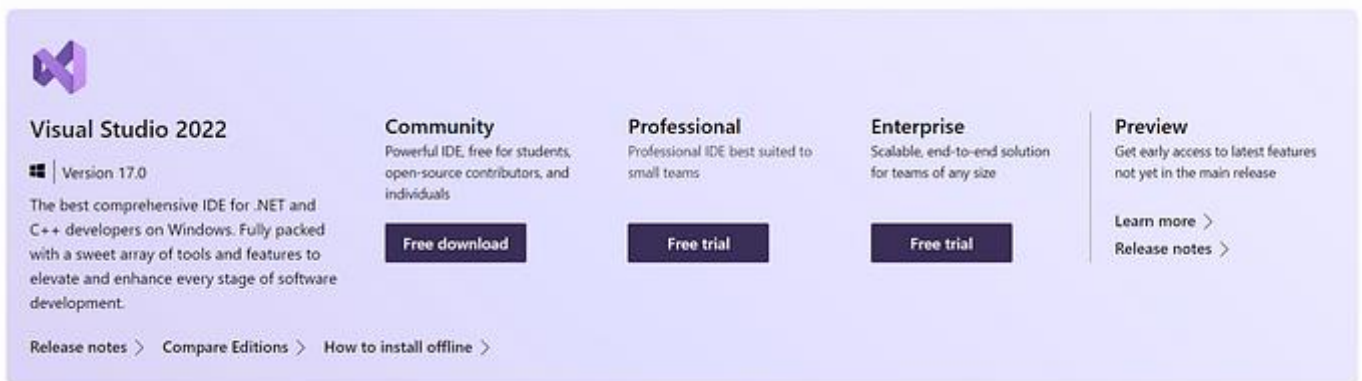
Version 17.0 of Visual Studio 2022 supports applications written with .Net 6.0. We have downloaded and installed Visual Studio 2022 Community Edition Version 17.0. The Community edition of Visual Studio 2022 is available for free.

So, let's start

Step 1

You can download Visual Studio 2022 version 17.0 from this [link](#).

Downloads



The screenshot shows the Visual Studio 2022 download page. It features the Visual Studio logo and the text 'Visual Studio 2022 | Version 17.0'. Below this, it states 'The best comprehensive IDE for .NET and C++ developers on Windows. Fully packed with a sweet array of tools and features to elevate and enhance every stage of software development.' There are four main sections: 'Community' (Powerful IDE, free for students, open-source contributors, and individuals) with a 'Free download' button; 'Professional' (Professional IDE best suited to small teams) with a 'Free trial' button; 'Enterprise' (Scalable, end-to-end solution for teams of any size) with a 'Free trial' button; and 'Preview' (Get early access to latest features not yet in the main release) with 'Learn more' and 'Release notes' links. At the bottom, there are links for 'Release notes', 'Compare Editions', and 'How to install offline'.

Step 2

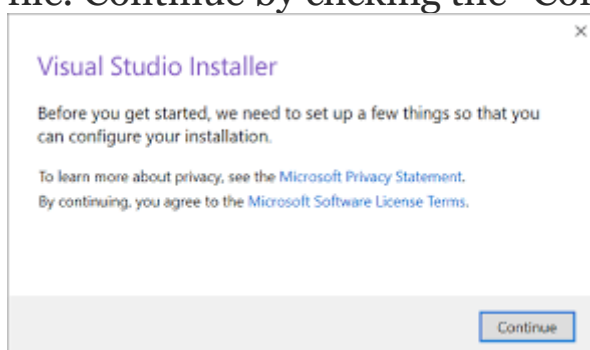
To download the VS 2022 executable file to the downloads directory, click the “Download” button.



The screenshot shows the Microsoft Visual Studio download page. At the top, there's a navigation bar with links for Microsoft, Visual Studio, Products, Downloads, Buy, Support, and Subscriber Access. A 'Free Visual Studio' button is also present. Below the navigation bar, a large purple banner reads 'Thank you for downloading Visual Studio' and 'Your download will start shortly. If your download does not begin, [click here to retry](#)'. On the left, there's a 'Getting started' dropdown menu and a 'Follow us' section with social media icons and a link to sign up for the developer community newsletter. The main content area has a section titled 'Install Visual Studio' with a sub-header 'Getting Started with Visual Studio IDE'. Below this, there's a video player showing 'Starting with Visual Studio' and a text block explaining that the Visual Studio IDE brings together graphical designers, compilers, code completion tools, source control, extensions, and many more features in one place.

Step 3

The Visual Studio installer window will open when you double-click the .exe file. Continue by clicking the “Continue” button.



Step 4

After hitting the Continue button, the download and installation progress bar window will appear.

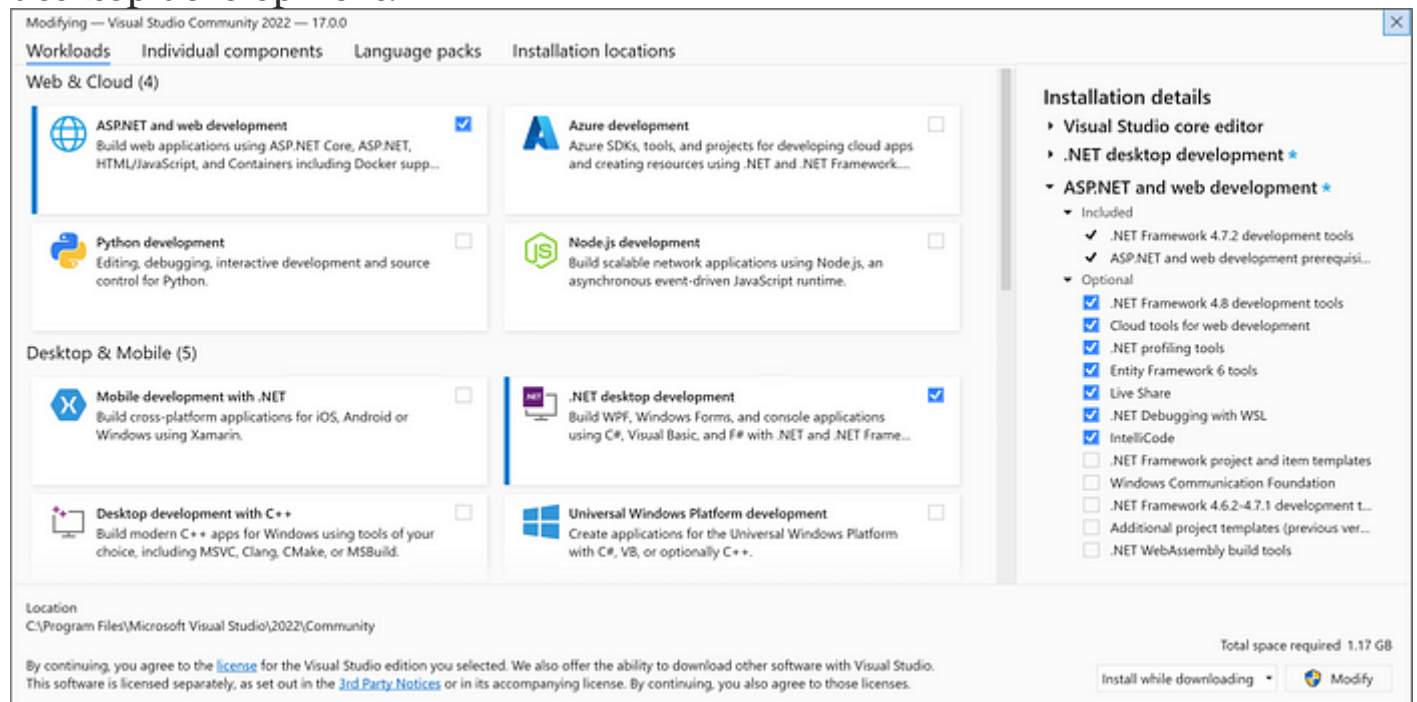
Visual Studio Installer

Almost done ... Getting everything ready.



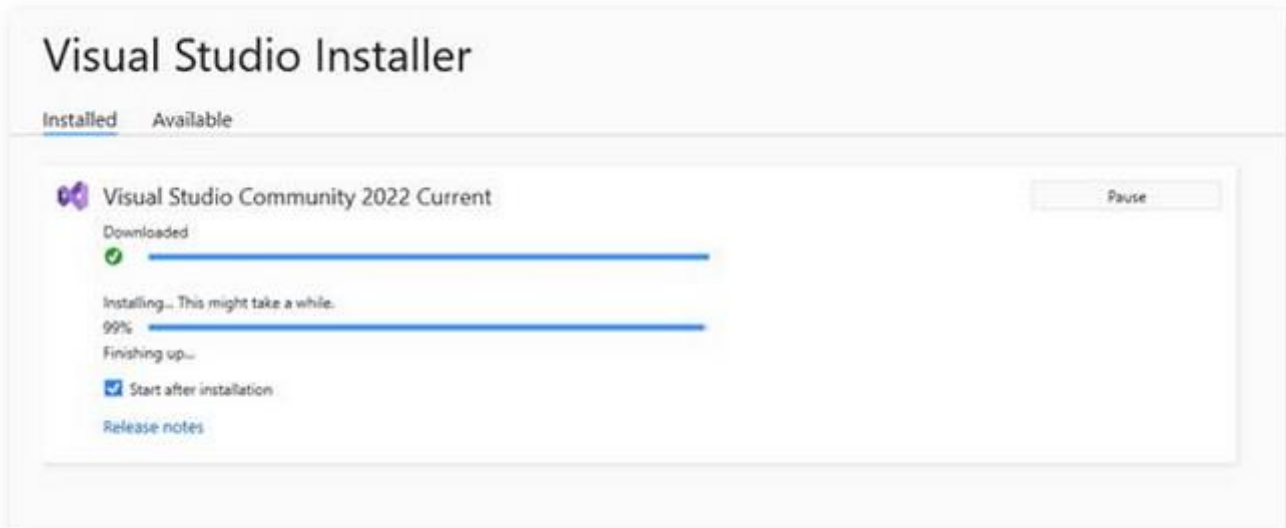
Step 5

Workloads will open after the download and installation are completed. We must decide which workloads we require. We chose ASP.NET and .NET desktop development.



Step 6

After selecting the appropriate packages, simply click the “Install” button.



When the installation is finished and we open Visual Studio for the first time, it takes 15–30 seconds for its initial internal setup.



We're getting things ready

Thanks for your patience.

.....

Your existing Visual Studio profile will be loaded. It may also request that you sign in to your Microsoft account. You can establish a new Microsoft account if you don't already have one.



The sign-in process is optional, so you can skip it. But don't forget to do this! Log up with your Gmail or Microsoft account to activate Visual Studio and receive a free lifetime licence.

If you don't have an account yet, select Not now, perhaps later.

Visual Studio

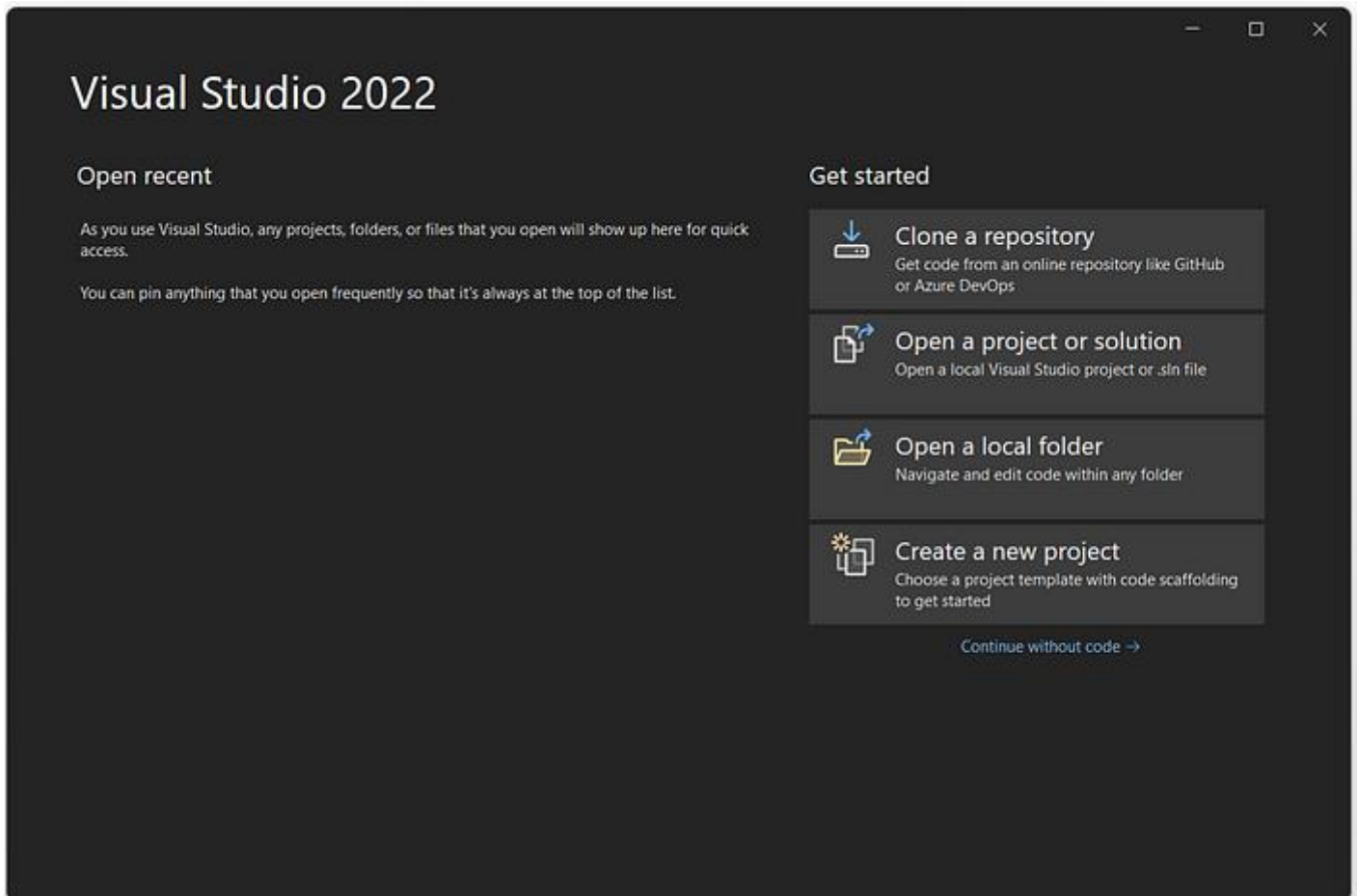
Sign in to Visual Studio

- Sync settings across devices
- Collaborate in real time with LiveShare
- Integrate seamlessly with Azure Services

Sign in

No account? [Create one!](#)

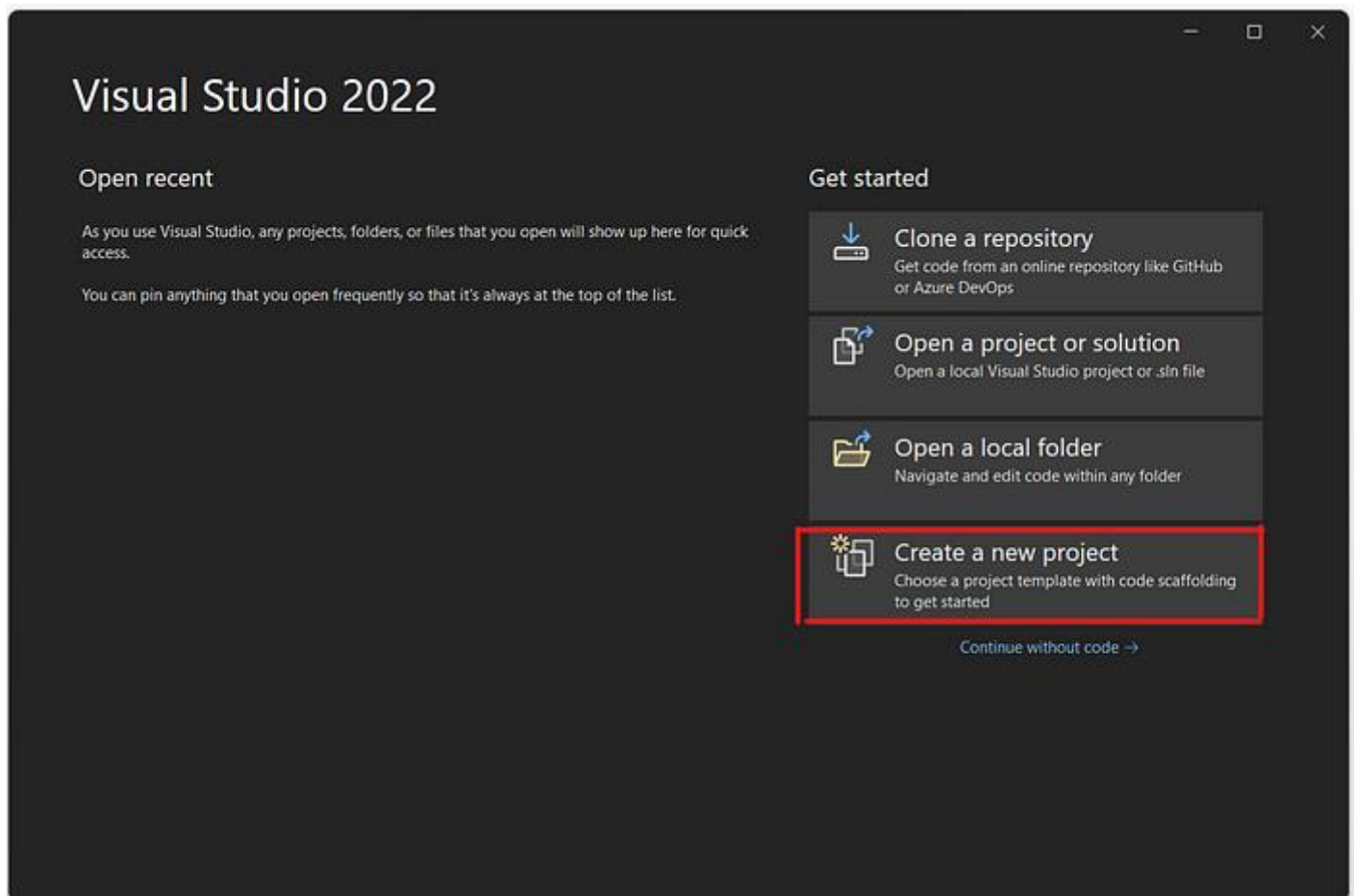
The installation of Visual Studio 2022 is now complete, and you can use it. We can observe a newly revamped Visual Studio landing page with a list of recent projects. We can also use GitHub and Azure DevOps to clone or check out the code. A project or solution, as well as a folder, can be opened. We can also start a new project.



Creating Your First Project

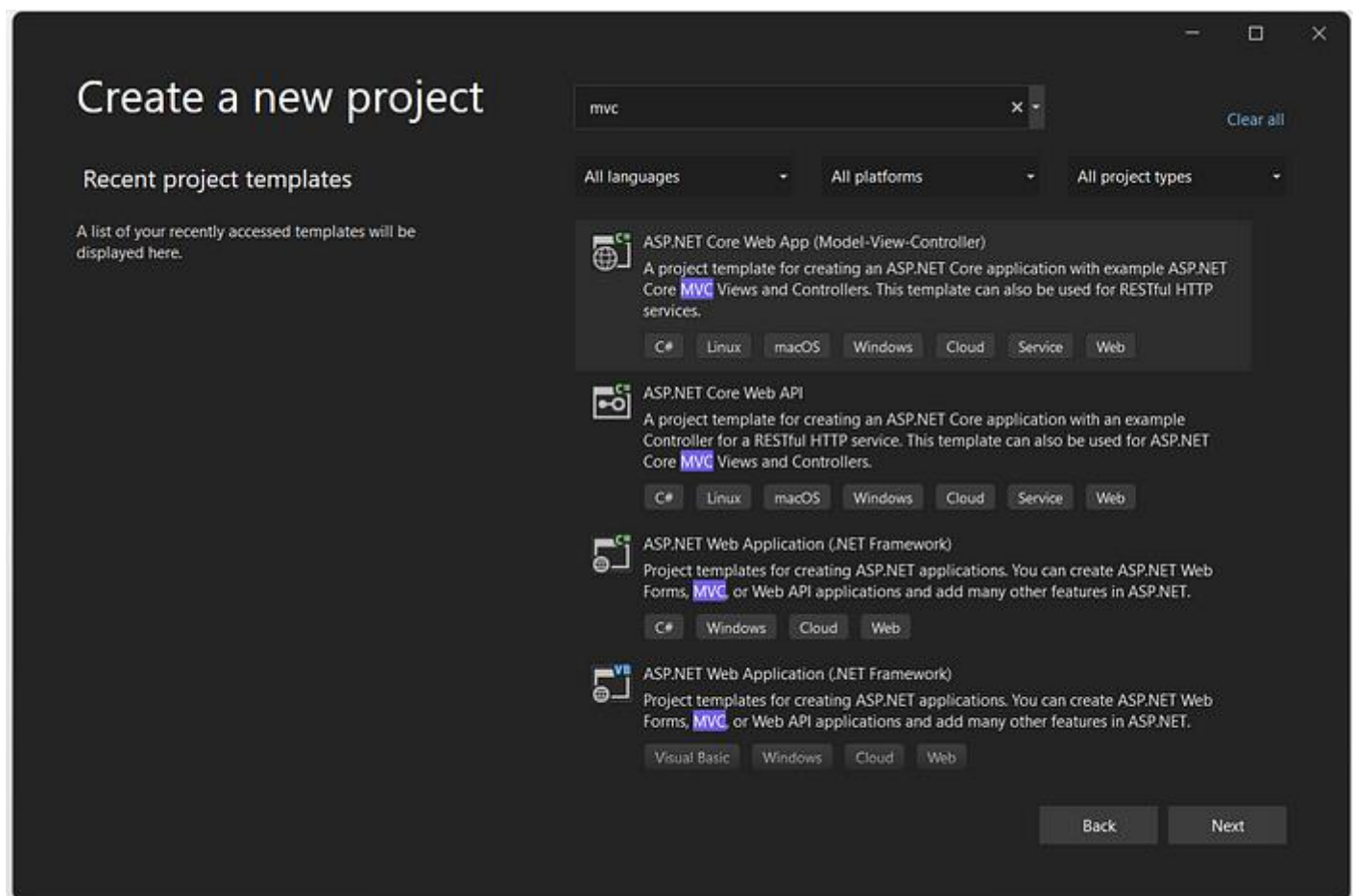
Step1

Click on Create a new Project to create a new project.

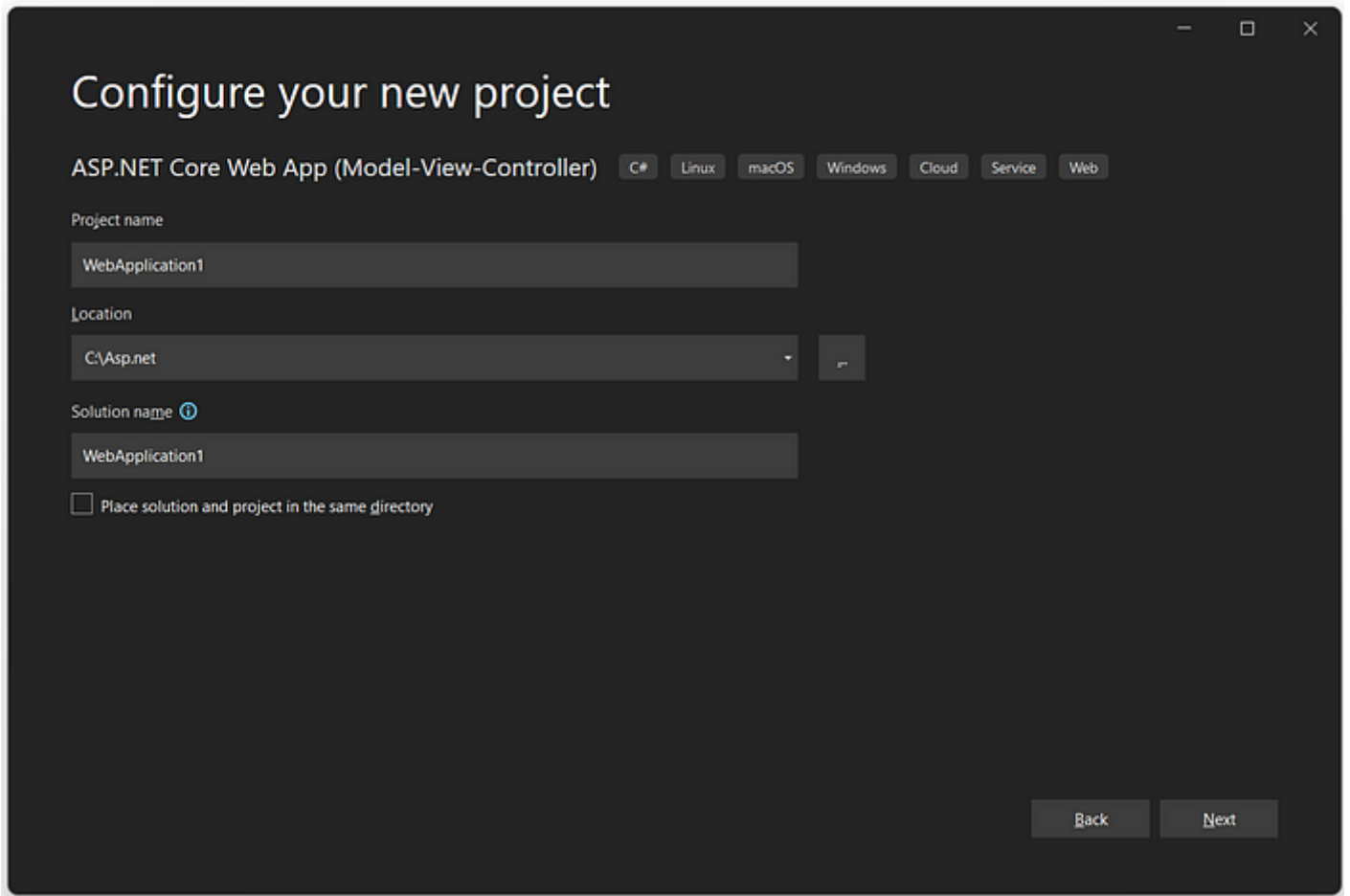


There are a variety of project types to choose from. For the time being, select the “Asp.Net Core MVC Web Application” project type.

Click Next after selecting Asp.Net Core MVC Application.



Give your project a meaningful name and select a path for it. Then click Next button.



Configure your new project

ASP.NET Core Web App (Model-View-Controller) C# Linux macOS Windows Cloud Service Web

Project name

WebApplication1

Location

C:\Asp.net

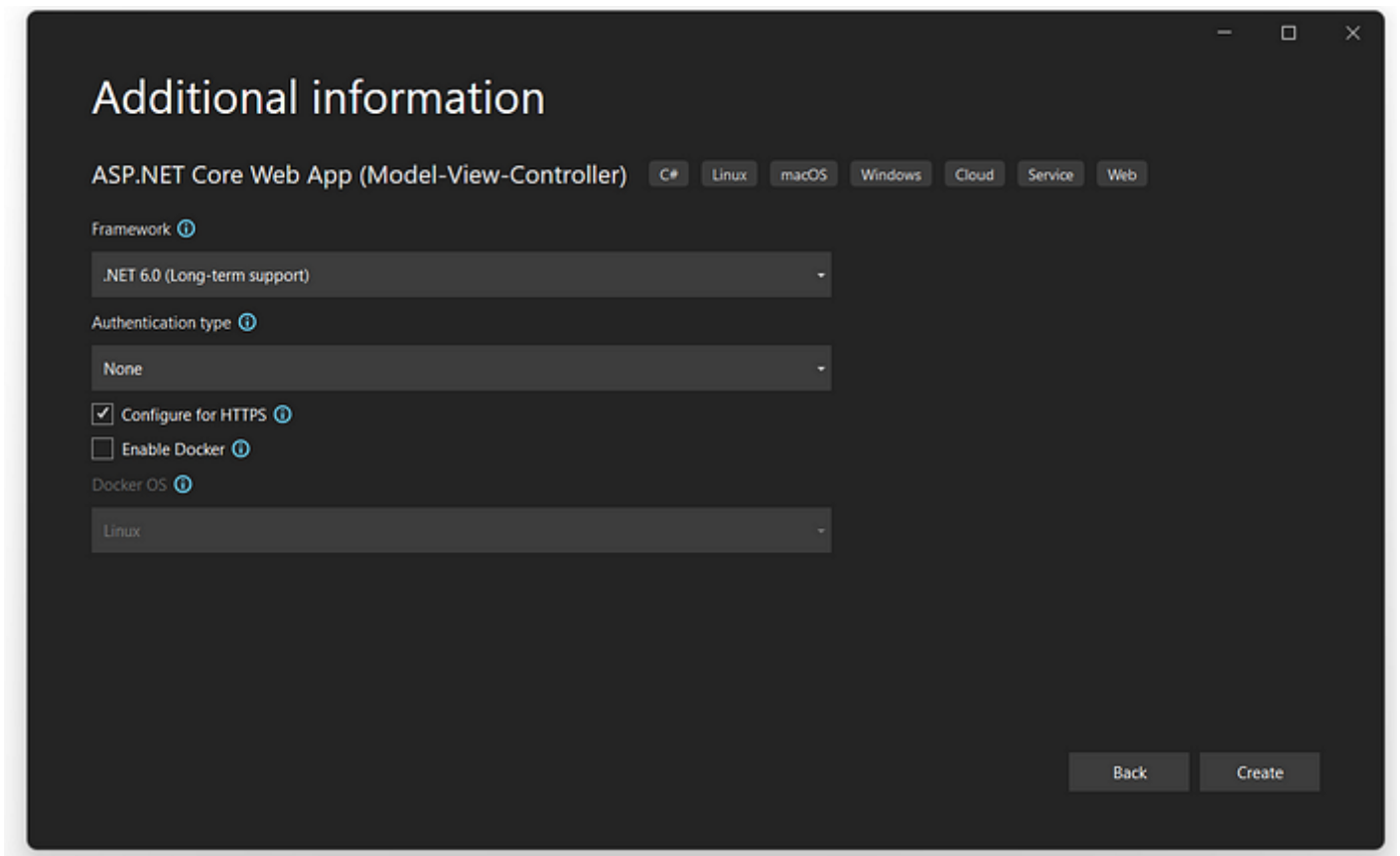
Solution name ⓘ

WebApplication1

☐ Place solution and project in the same directory

Back Next

Now, choose framework .NET 6.0. Then click the create button.



Additional information

ASP.NET Core Web App (Model-View-Controller) C# Linux macOS Windows Cloud Service Web

Framework ⓘ

.NET 6.0 (Long-term support)

Authentication type ⓘ

None

☒ Configure for HTTPS ⓘ

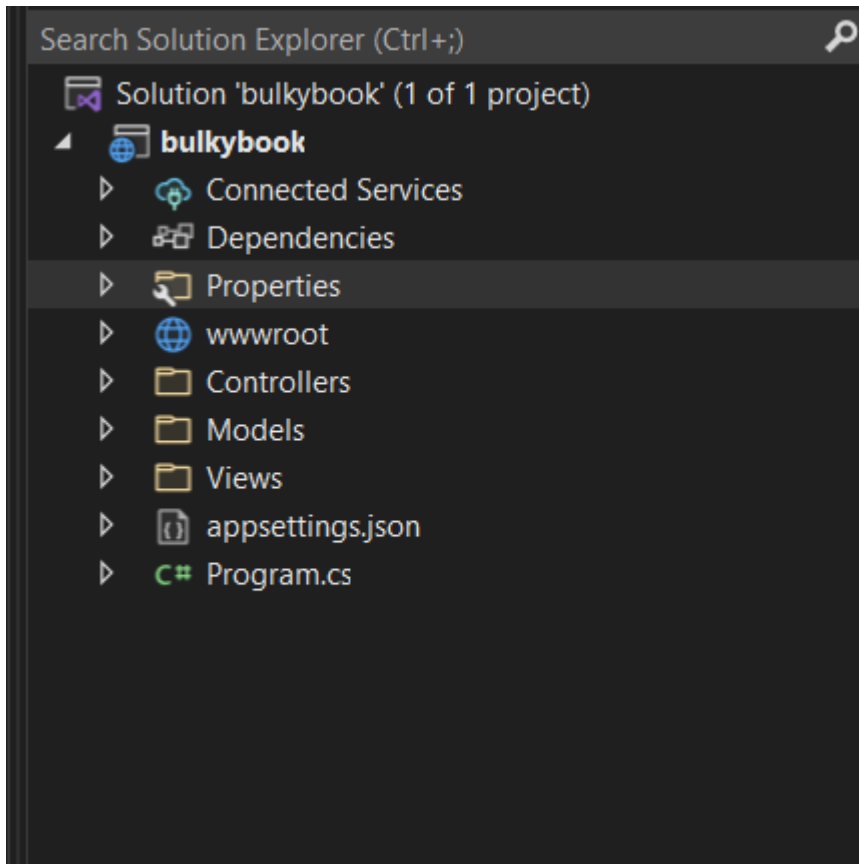
☐ Enable Docker ⓘ

Docker OS ⓘ

Linux

Back Create

Asp.Net Core MVC 6.0 application created and project structure is shown below,



Now, build and run the application.

OUTPUT

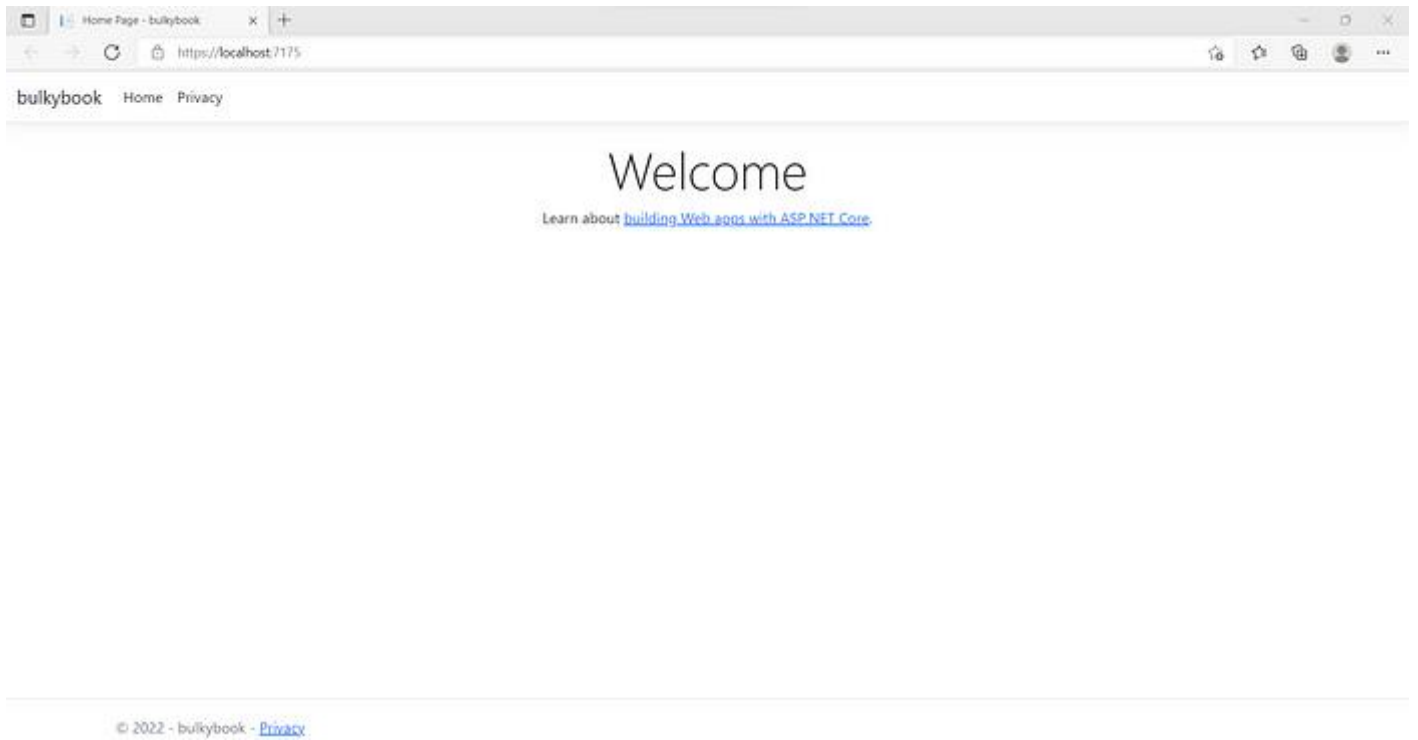


CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY



Tool Base Manual || M.Sc. (IT) || SEM II || MS226 : Http Web Service For Enterprise Application



Tool Based Manual

Installation of Apache, MySQL and PHP

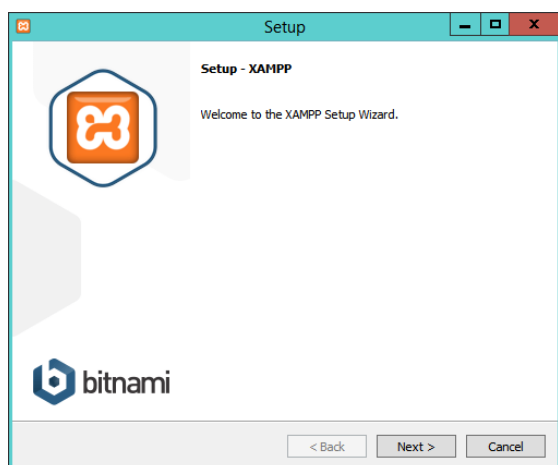
- If you wish to develop Internet applications but don't have your own development server, you can develop one such on your local computer.
- So, testing can be as easy as saving an update (usually just a matter of clicking once on an icon) and then hitting the Refresh button in your browser.
- Following are the fully functioning setup used for developing dynamic Internet web pages.
 1. WAMP - Windows, Apache, MySQL, and PHP
 2. MAMP - Mac, Apache, MySQL, and PHP
 3. LAMP - Linux, Apache, MySQL, and PHP
- WAMPs, MAMPs, and LAMPs come in the form of a package that binds the bundled programs together so that you don't have to install and set them up separately.
- This means you can simply download and install a single program, and follow a few easy prompts, to get your web development server up and running in the quickest time with a minimum hassle.

Installing Xampp on windows

- There are several available WAMP servers, each offering slightly different configurations, but out of the various open source and free options, the best is probably XAMPP.

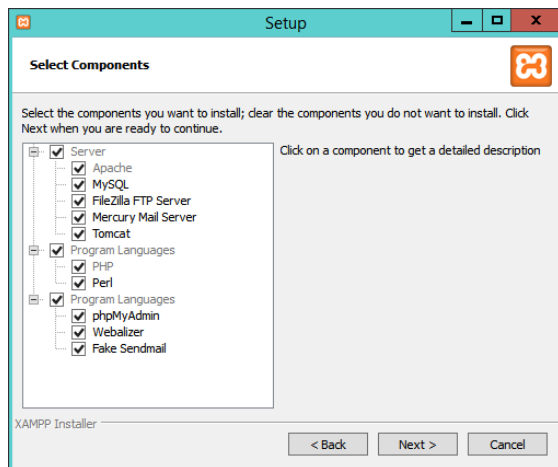
Link: <http://apachefriends.org>

Installation Steps

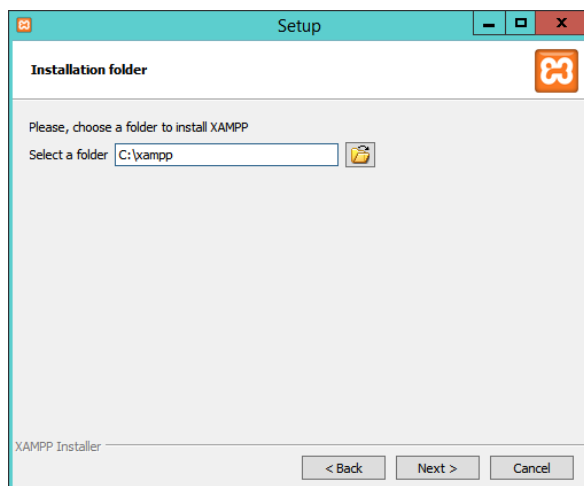


Once you've downloaded the installer, run it. It will show the following window.

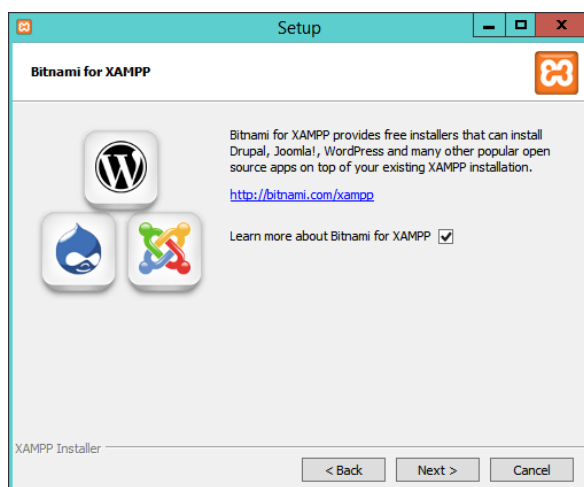
Tool Base Manual || M.Sc. (IT) || SEM II || MS227 : Web Development Using Open Source



Uncheck any of the components that you don't need. For example, at the minimum for this book you will want to keep Apache, MySQL, PHP, and PHPMyAdmin checked.

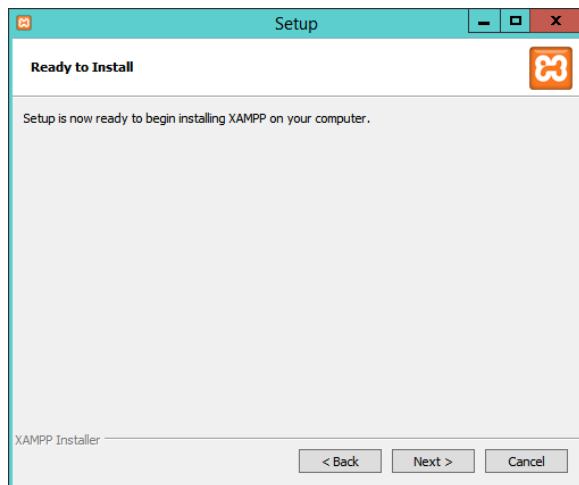


Choose a folder for the installation. You are recommended to accept the default provided unless you have a good reason to choose a different one. If the folder you choose exists and is not empty, you will be unable to use it.

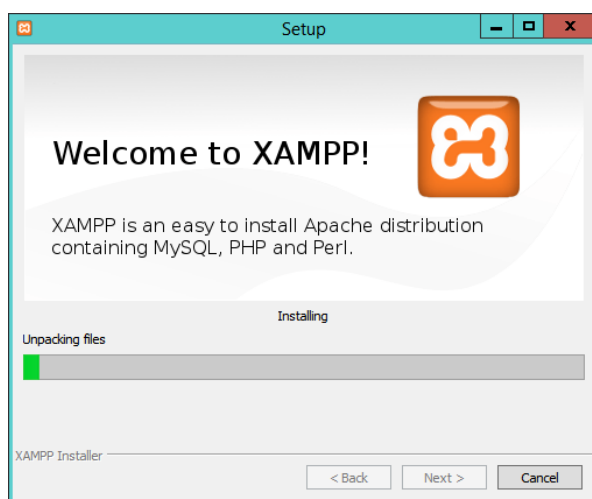


The screen is for supplying information on adding free installers for related products in a new web browser window or tab.

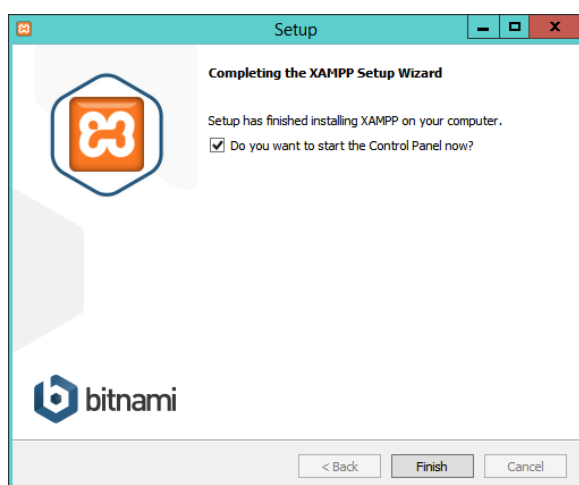
Tool Base Manual || M.Sc. (IT) || SEM II || MS227 : Web Development Using Open Source



Setup is ready to start. Click next

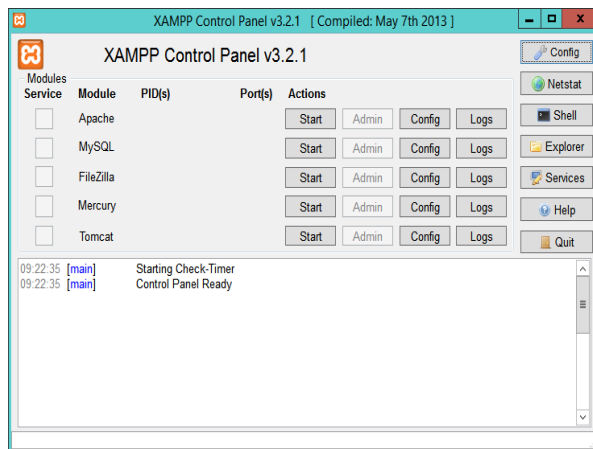


Installation screen. During installation, you can click any of the icons to open up a web browser window displaying information on the product shown. The whole process should take only a few minutes on most computers.



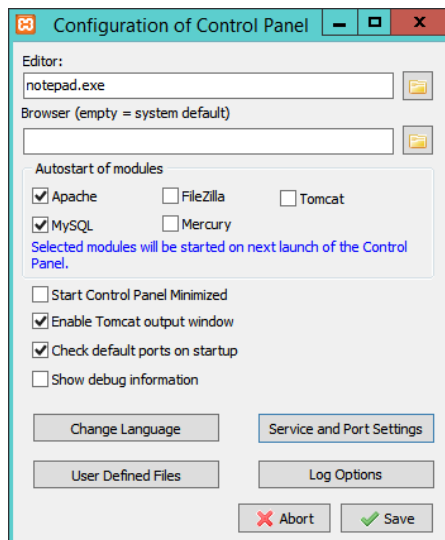
Once installation is complete, the screen will be displayed with a check box already checked for starting the XAMPP control panel. Click Finish.

Tool Base Manual || M.Sc. (IT) || SEM II || MS227 : Web Development Using Open Source



Now you are ready to start using XAMPP by configuring it from the control panel.

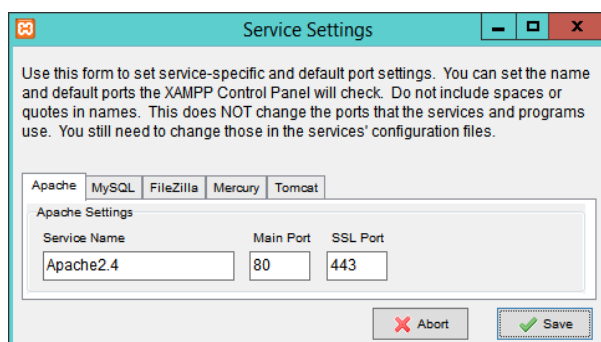
Click the Config button at the top-right corner



Check the Apache and MySQL boxes to ensure they auto start.

OR

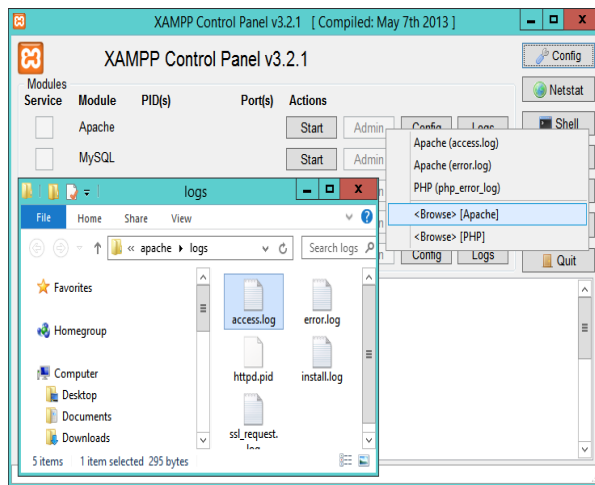
You can simply click the Start buttons for Apache and MySQL to have them run for this session only.



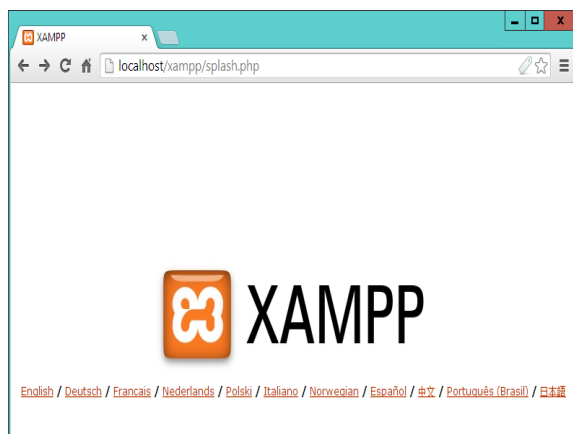
Alter the port settings used by clicking Service and Port Settings to call up the window

The default ports assigned will normally be port 80 for the Apache web server, 443 for SSL, and 3306 for MySQL.

Tool Base Manual || M.Sc. (IT) || SEM II || MS227 : Web Development Using Open Source



If the Apache Log button has been clicked, and the folder of log files opened.



Browse in your default browser
Localhost Or 127.0.0.1

Port number if you changed
localhost:<port>
127.0.0.1:<port>

Tool Based Manual

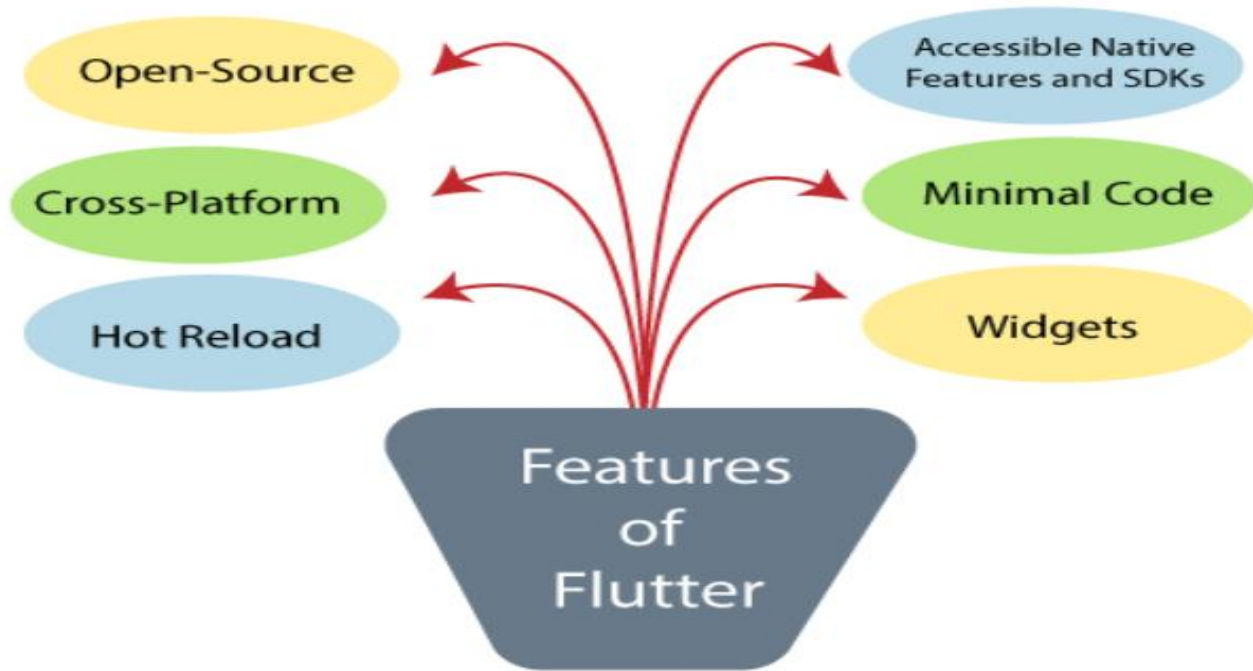
Flutter:

- Mobile UI framework for creating native apps for iOS & Android
- Flutter is a single codebase framework for application development
- Flutter is a tool that allows to build native cross-platform (Android, iOS, Web, Desktop) applications in one programming language and codebase.
- Single-Code base(dart) means only we have to write our app once for multiple devices.
- Cross-platform is a system that works across multiple types of operating systems. Cross-platform is known as multiplatform or platform independent.
- Flutter is Google's Mobile SDK to build native iOS and Android apps from a single codebase.

Cross-Platform, Native & Hybrid

- **Cross- Platform:** Cross-platform is a system that works across multiple types of operating systems. Cross-platform is known as multiplatform or platform independent. Write code once and reuse for different platforms.
- **Native :** An application developed exclusively for a single platform using a native-to-the-operating – system language is a Native app. (Android- Java/Kotlin, UI-XML-In Android Studio).
- The process of developing an application using Java/Kotlin and for UI using XML in Android Studio where the logic part is developed by Java or Kotlin. The application developed in core languages and deployed on the specific platform then it is known as native application.
- **Hybrid :** A blend solution that consists of both native and web solutions. Developers embed the code written in web development (HTML, JavaScript, CSS) into native app using plugins Apache Cordova or Ionic Capacitor.





Installation & Set Up:

- Check for Flutter.dev on browser and click on get started

1. Download the flutter SDK

- Extract the flutter sdk
- Create a separate folder src in C drive and copy the extracted flutter file to src folder

2. Setup environment variables: We have to set up the env variable because we have to run Flutter Doctor on our command line.

- Go to windows search for env and edit the system environment variable
- Download Git and install at C:\Program Files
- Create a path variable and name the variable as path and values as C:\src\flutter\bin
- Find path variable in system and edit the path and create new path as
C:\src\flutter\bin
- C:\Program Files\Git\cmd in system variables
- C:\Windows\System32\WindowsPowerShell\v1.0 in system variables
- C:\Windows\System32 in system variables



Installation & Set Up:

Then write flutter in cmd and check the screen like this:

```
cmd Command Prompt
Microsoft Windows [Version 10.0.19044.2251]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CHARUSAT>flutter
Manage your Flutter app development.

Common commands:

  flutter create <output directory>
    Create a new Flutter project in the specified directory.

  flutter run [options]
    Run your Flutter application on an attached device or in an emulator.

Usage: flutter <command> [arguments]

Global options:
-h, --help            Print this usage information.
-v, --verbose         Noisy logging, including all shell commands executed.
                       If used with "--help", shows hidden options. If used with "flutter doctor", shows additional
                       those cases.)
-d, --device-id       Target device id or name (prefixes allowed).
--version             Reports the version of this tool.
--suppress-analytics  Suppress analytics reporting when this command runs.

Available commands:

Flutter SDK
bash-completion  Output command line shell completion setup scripts.
channel          List or switch Flutter channels.
config           Configure Flutter settings.
doctor          Show information about the installed tooling.
downgrade       Downgrade Flutter to the last active version for the current channel.
precache        Populate the Flutter tool's cache of binary artifacts.
upgrade         Upgrade your copy of Flutter.

Project
analyze         Analyze the project's Dart code.
assemble       Assemble and build Flutter resources.
build          Build an executable app or install bundle.
clean          Delete the build/ and .dart_tool/ directories.
create         Create a new Flutter project.
drive          Run integration tests for the project on an attached device or emulator.
format         Format one or more Dart files.
gen-l10n       Generate localizations for the current project.
pub            Commands for managing Flutter packages.
run            Run your Flutter app on an attached device.
test           Run Flutter unit tests for the current project.
```

```
Tools & Devices
attach         Attach to a running app.
custom-devices List, reset, add and delete custom devices.
devices        List all connected devices.
emulators      List, launch and create emulators.
install        Install a Flutter app on an attached device.
logs           Show log output for running Flutter apps.
screenshot     Take a screenshot from a connected device.
symbolize      Symbolize a stack trace from an AOT-compiled Flutter app.

Run "flutter help <command>" for more information about a command.
Run "flutter help -v" for verbose help output, including less commonly used options.
```

Welcome to Flutter! - <https://flutter.dev>

The Flutter tool uses Google Analytics to anonymously report feature usage statistics and basic crash reports. This data is used to help improve Flutter tools over time.

Flutter tool analytics are not sent on the very first run. To disable reporting, type 'flutter config --no-analytics'. To display the current setting, type 'flutter config'. If you opt out of analytics, an opt-out event will be sent, and then no further information will be sent by the Flutter tool.

By downloading the Flutter SDK, you agree to the Google Terms of Service. Note: The Google Privacy Policy describes how data is handled in this service.

Moreover, Flutter includes the Dart SDK, which may send usage metrics and crash reports to Google.

Read about data we send with crash reports:
<https://flutter.dev/docs/reference/crash-reporting>

See Google's privacy policy:
<https://policies.google.com/privacy>

```
C:\Users\CHARUSAT>flutter doctor
```



Installation & Set Up:

Then Write flutter doctor on cmd and check the screen as : This tells which tools are missing in flutter development

```
C:\Users\CHARUSAT>flutter doctor
Running "flutter pub get" in flutter_tools... 18.1s
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.3.9, on Microsoft Windows [Version 10.0.19044.2251], locale en-US)
[!] Android toolchain - develop for Android devices (Android SDK version 33.0.1)
    X cmdline-tools component is missing
      Run `path/to/sdkmanager --install "cmdline-tools;latest"`
      See https://developer.android.com/studio/command-line for more details.
    X Android license status unknown.
      Run `flutter doctor --android-licenses` to accept the SDK licenses.
      See https://flutter.dev/docs/get-started/install/windows#android-setup for more details.
[✓] Chrome - develop for the web
[X] Visual Studio - develop for Windows
    X Visual Studio not installed; this is necessary for Windows development.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its default components
[✓] Android Studio (version 2021.3)
[✓] Connected device (3 available)
[✓] HTTP Host Availability

! Doctor found issues in 2 categories.

C:\Users\CHARUSAT>
```

3. Install Android Studio

- Open Android SDK Manager from tools in android and then click on Android SDK Command-line tools latest and download the same.
- Then go to cmd and run : `flutter doctor --android-licenses` and then write yes for all licenses
- Then go to cmd and run flutter doctor again

Command Prompt

```
ase Materials may not be fully tested and may contain bugs or errors; (ii) Pre-Release  
mmercial release in their current state; (iii) regulatory approvals for Pre-Release  
ot been obtained, and Pre-Release Materials may therefore not be certified for use i  
or may not be suitable for certain applications and (iv) MIPS can provide no assuran  
generally available a production version of the Pre-Release Materials . MIPS is not  
/or release or offer for sale or license a final product based upon the Pre-Release  
to abandon the Pre-Release Materials or any such development platform at any time a  
ty whatsoever to Recipient or any other person.
```

```
ANY PRE-RELEASE MATERIALS ARE NON-QUALIFIED AND, AS SUCH, ARE PROVIDED AS IS AND AS  
WITHOUT REPRESENTATION OR WARRANTY OF ANY KIND.
```

```
10.8 Open Source Software. In the event Open Source software is included with Evalua  
ware is licensed pursuant to the applicable Open Source software license agreement i  
e comments in the applicable source code file(s) and/or file header as indicated in  
detail may be available (where applicable) in the accompanying on-line documentation  
ftware, nothing in this Agreement limits any rights under, or grants rights that sup  
Open Source software license agreement.
```

```
-----  
Accept? (y/N): y  
All SDK package licenses accepted
```

```
C:\Users\CHARUSAT>
```

3.1 Install flutter plugins from File -Setting-Plugins-Flutter- Install flutter

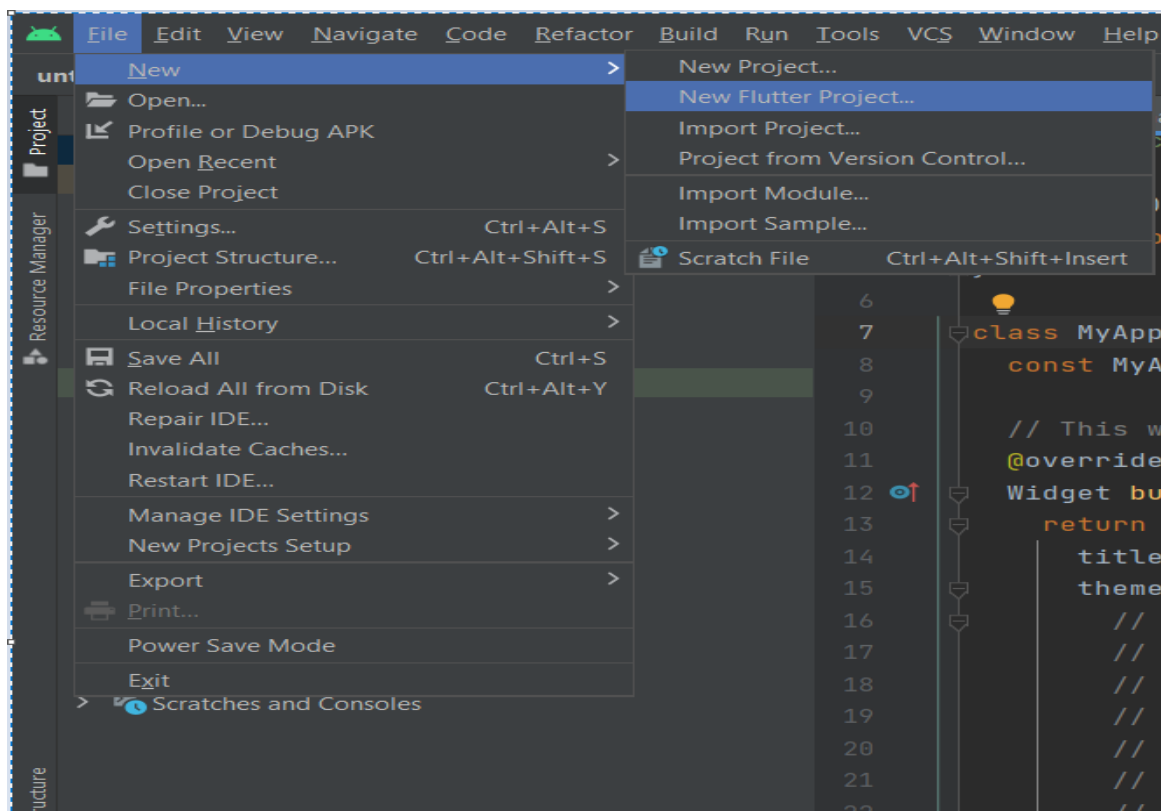
3.2 Then install dart in the plugins in setting panel.

4. First go to the file location C:\KKFlutter-2022, then run flutter command and write the command flutter create kkfirstapp and hit enter, it will create the flutter project in the directory. Then to open the project click open project and select the directory and choose the app name where the project is created and the project is ready.
5. Lets see how to create flutter project in Android studio: Go to File- Open -New Flutter Project, choose the flutter sdk path and click next, Provide the name of the Project, Choose the project location, Give project description, project type- Application, choose android language: Kotlin, iOS – Swift, Choose Platform: Android, iOS, Web and then click on finish.
6. Install Flutter and Dart Plugins

7. Agree to all licenses

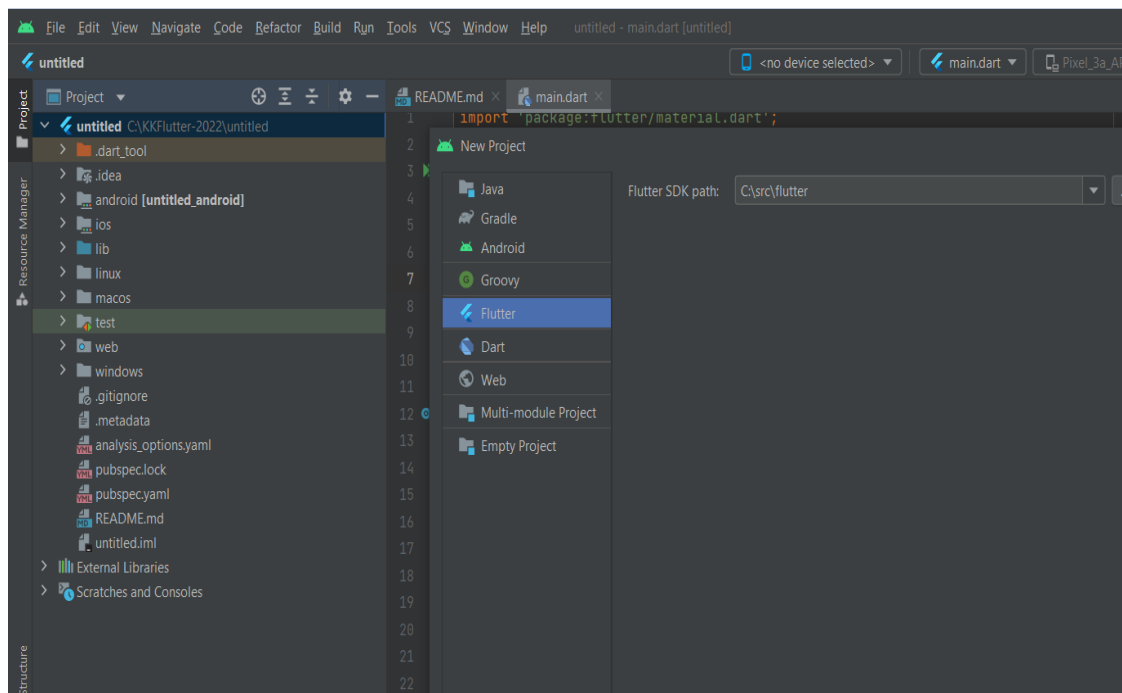
Demo Flutter project using Android Studio : Can be created on CMD & in Android Studio

1. Create a folder in any drive and specify the name of the folder where you can save the flutter project.
2. Go to Android Studio -> File-> New Flutter Project

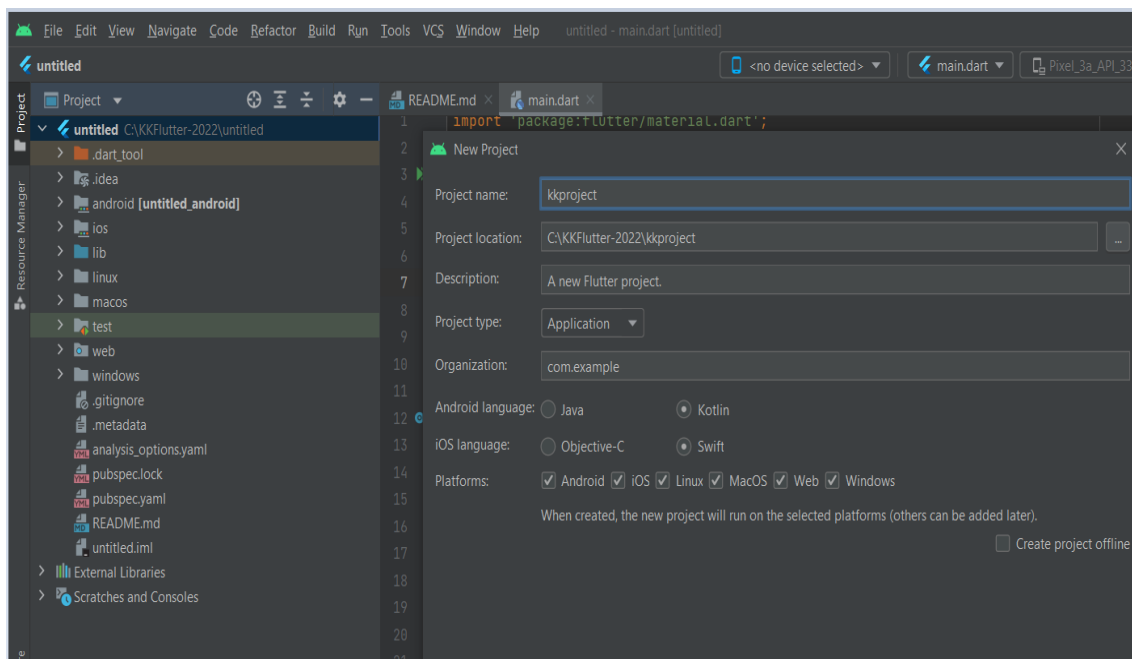


3. Choose Flutter SDK location and click next

Tool Based Manual || MCA || SEM II || CA871 : Software Engineering with Agile and DevOps

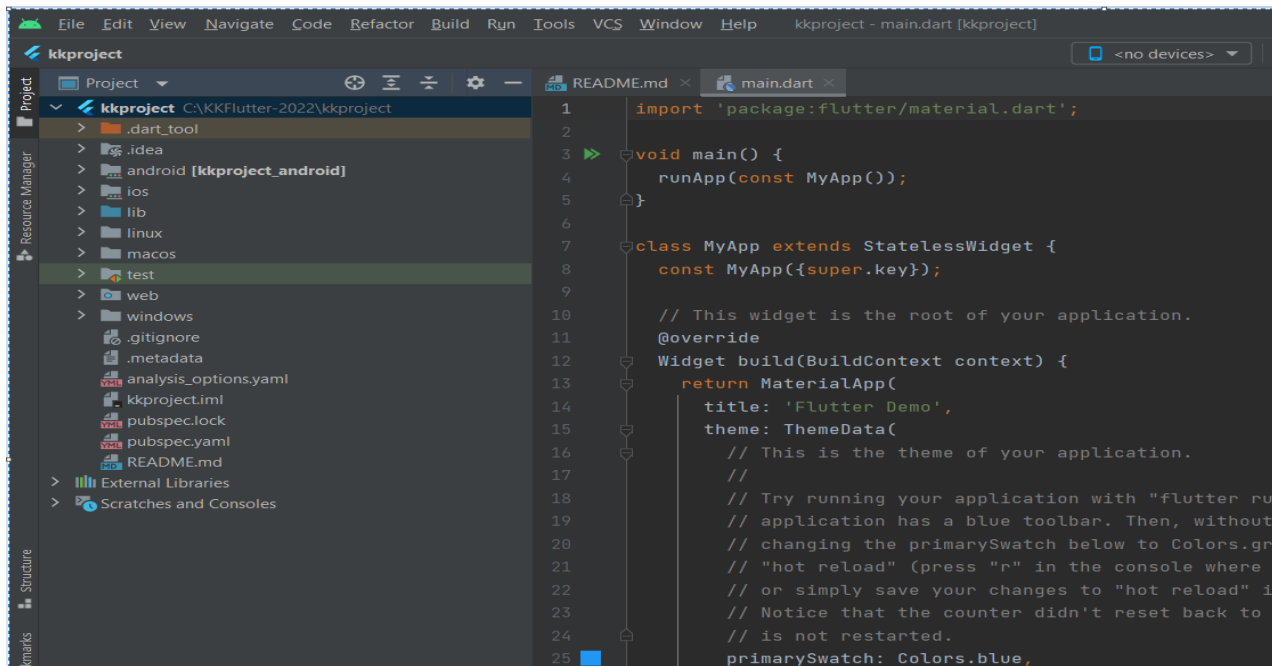


4. Provide the name of Project in small letter-> specify the project location->Write the description of project -> Choose Application type as : Application -> Choose Android Language -> Choose iOS Language -> Choose the Platform for which you want to develop the project then click on Finish.



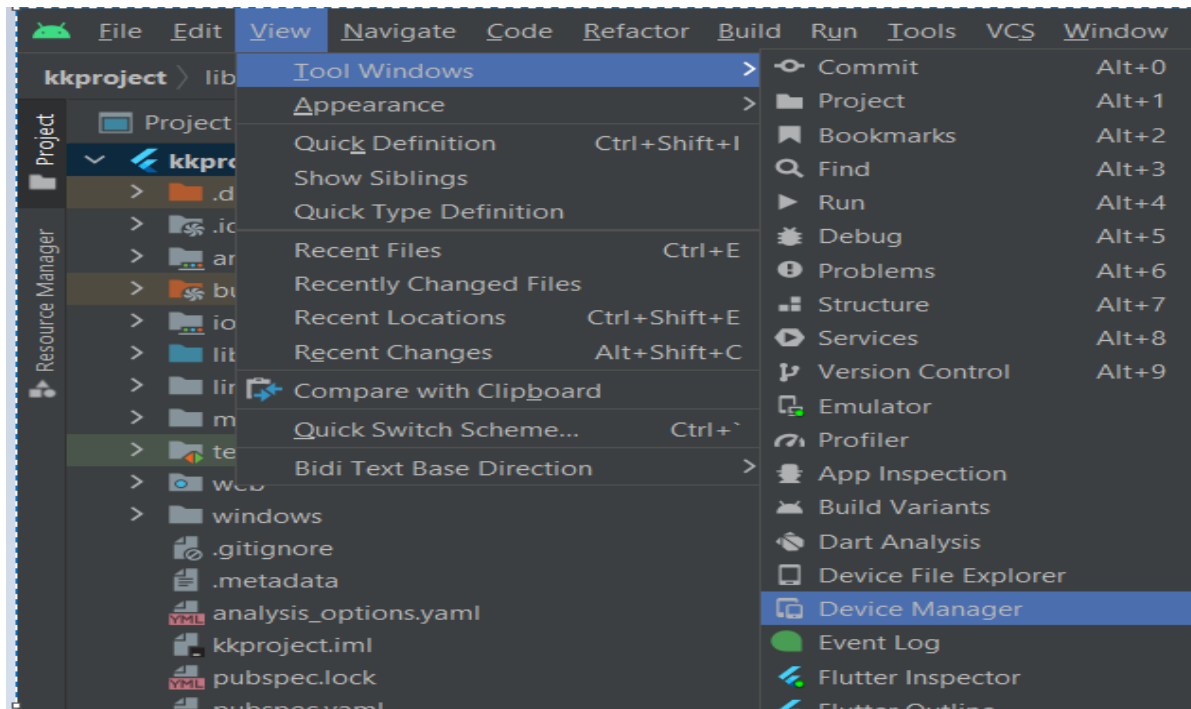
5. A new project window will appear on the screen

Tool Based Manual || MCA || SEM II || CA871 : Software Engineering with Agile and DevOps

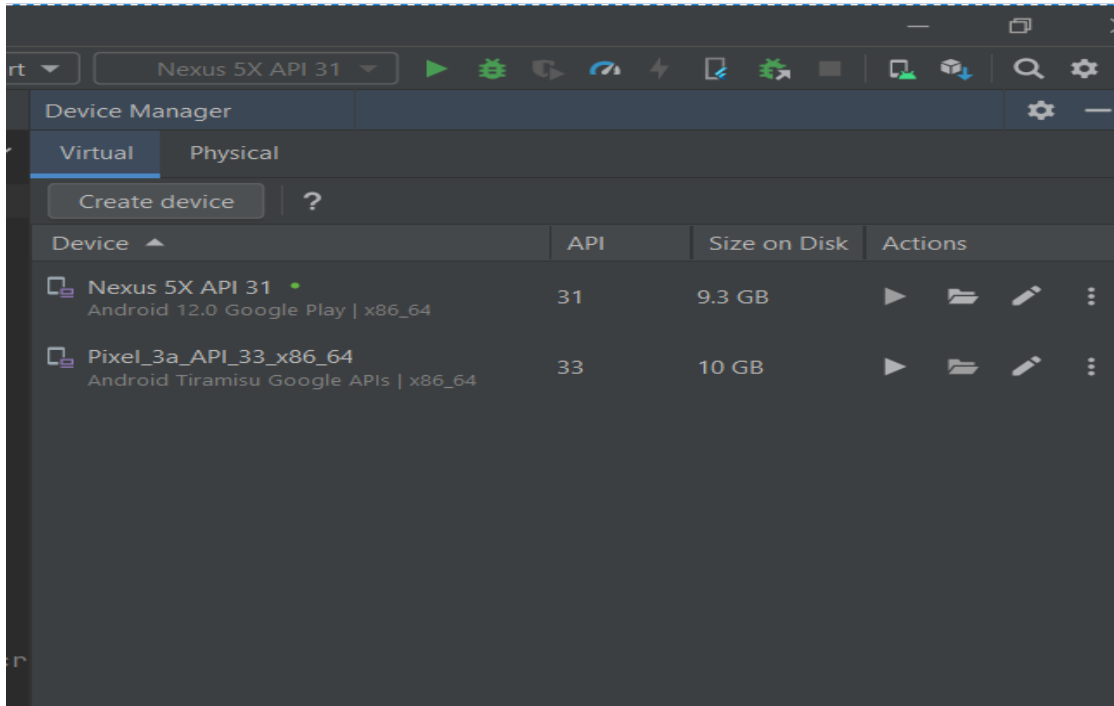


Configuring Android Virtual Device:

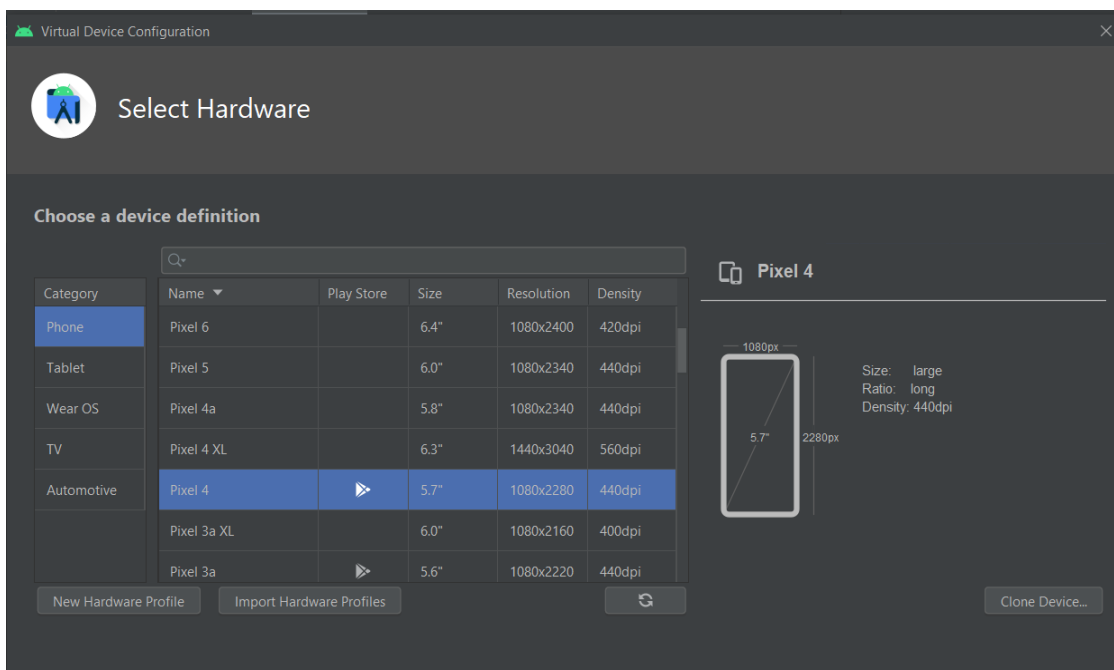
1. Go to View -> Tool Window -> Device Manager



2. Create Virtual Device



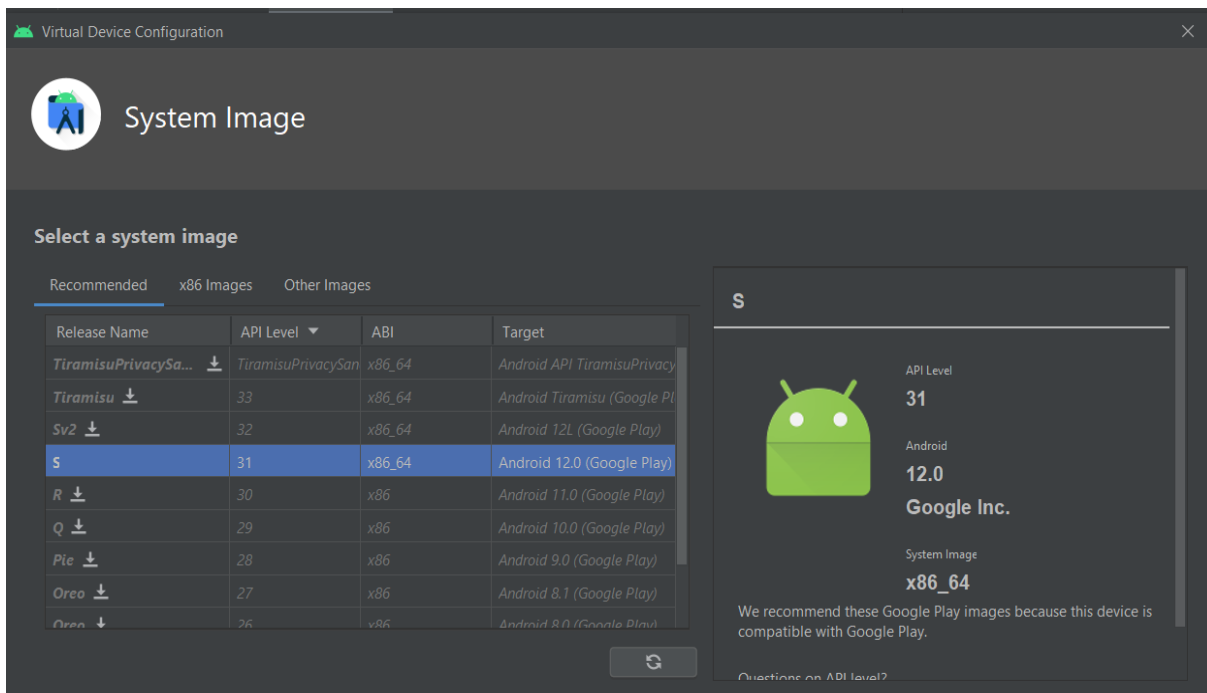
3. Choose Device



4. Choose OS



Tool Based Manual || MCA || SEM II || CA871 : Software Engineering with Agile and DevOps



Set Name of AVD and click finish

